

2. a) Properties of an ideal switch

- There is no limit on the amount of current in either direction.
- No limit on operating speed of the wire when it changes state (zero rise and fall times).
- Uses no power when in operation.
- Zero resistance and zero voltage drop when closed switch ($R = 0\Omega$ and $V = 0V$).
- Infinite resistance and zero current when open switch ($R = \infty\Omega$ and $I = 0A$).
- Operates over all bandwidths and frequencies.
- Has zero leakage current.

2. b) Properties of a real switch:

- **Limited current rating**
 - Every real switch has a maximum current rating.
 - Exceeding the rating can result in permanent damage and overheating.
 - Common mechanical and MOSFET switches have current ratings of 1A-5A and 5A-30A respectively.
- **Limited voltage range**
 - Every real switch has a maximum voltage bound.
 - Exceeding the rating can result in clipping of voltage and possible permanent damage.
 - The output voltage from the switch will always be a little less than the input.
 - $V_{out} \leq V_{supply} - V_{sat}$
 - V_{sat} depends on MOSFET synthesis and environmental factors such as temperature.
- **Non-zero operating/switching speed**
 - There exists a small time difference between rise and fall times.
 - This can cause propagation delays in other components.
 - $t_{ON} = t_d + t_r$, where t_{ON} is turn-on time, t_d is delay time, and t_r is rise time.
 - $t_{OFF} = t_s + t_f$, where t_{OFF} is turn-off time, t_s is storage time, and t_f is fall time.
 - The rise and fall times are proportional to the time constant caused by a load capacitance and the ON or OFF resistance of the switch.
 - $t_r \approx 2.2R_{ON}C_L$
 - $t_f \approx 2.2R_{OFF}C_L$
- **Non-zero closed switch resistance and voltage**
 - Practical switch has an associated resistance with it when closed.
 - This causes a voltage drop across it.

- Power is dissipated through it as heat.
- $V_{drop} = I \times R_{ON}$
- $P_{abs} = I^2 \times R_{ON}$
- **Limited open switch resistance and leakage current**
 - Practical switch has an associated current flowing through it even when it is open.
 - Output wire is not perfectly insulated.
 - $I_{leak} = \frac{V_{drop}}{R_{OFF}}$
- **Operates best over certain bandwidth**
 - Due to internal capacitances between terminals, certain frequencies get attenuated.
 - Hence input bandwidth and frequencies are limited.
 - $f_{min} = \frac{1}{2\pi R_{ON} C_{interanl}}$
 - Frequencies higher than f_{min} will get significantly cut off with lower and equal frequencies getting no significant change.
- **Voltage spikes due to charge injection**
 - Switching between the ON and OFF states causes stored charge in the MOSFETS to be injected into the output.
 - This causes small voltage spikes
 - This spike in voltage depends on the amount of charge injected and load capacitances.
 - $\Delta V = \frac{Q_{inj}}{C_{load}}$

References:

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